VII. A brief introduction to local fields

1 Topology and valuations (Now we assume K is a local field, $k = \mathcal{O}_K/\mathfrak{m}_K$ be the residue field)

Def VII.1.1. A local field is a field K with a nontrivial absolute value $|\cdot|$ s.t. K is locally compact

Prop VII.1.2. Let K complete (with discrete absolute value). $A = \mathcal{O}_K$ and $\mathfrak{m}$ be maximal in A.

Then A compact $\Leftrightarrow$ $A/\mathfrak{m}_K$ finite.

Pf: $\Rightarrow$: $A/\mathfrak{m} = \bigsqcup_{s\in S}(s+\mathfrak{m})$ (S is the set of representatives for $A/\mathfrak{m}$), $\mathfrak{m} = \{x\in A \mid |x|<1\}$. open

$\Leftarrow$: $\forall a\in A$, $a = s_0 + s_1\pi + \cdots + s_n\pi^n + \cdots$ ($s_i\in S$). $|S|<\infty$. $\Rightarrow$ A is totally bounded, complete (closed)

$\Rightarrow$ A compact $\square$

Thm VII.1.3. (Classification of local fields). Let K be a local field, then we have

1°. If K is non-archimedean, $p = \operatorname{char} k$, (1). If $\operatorname{Char} K = 0$, then $\mathbb{Q}_p$ embeds into K with $e = v(p)$, $f = [k:\mathbb{F}_p]$ and $[K:\mathbb{Q}_p] = ef$;

(2) If $\operatorname{char} K = p$, then $K \cong \mathbb{F}_q((T))$, $q = |k| = p^f$. as topological fields. Equivalently $\mathcal{O}_K \cong \mathbb{F}_q[[T]]$

Rmk. Recall that $\mathcal{O}_K = \{x\in K \mid v(x)\geq 0\}$, $\mathfrak{m}_K = \{x\in K \mid v(x)\geq 1\}$, π is the uniformizer.

2°. If K is archimedean, then $K\cong\mathbb{R}$ or $K\cong\mathbb{C}$ (By Ostrowski's thm).

We need three preliminary lemmas:

Lemma VII.1.4. Let $S\subseteq\mathcal{O}_K$ be a set of representatives of $k = \mathcal{O}_K/\mathfrak{m}_K$ with $0\in S$, so that $S\to k$ is a bijec

Then $\forall\, x\in\mathcal{O}_K$, $\exists!$ convergent expansion $x = \sum_{n\geq 0} s_n\pi^n$. $s_n\in S$

$\forall\, x\in K$, $\exists N\in\mathbb{Z}$, $\exists!$ expansion $x = \sum_{n\geq N} s_n\pi^n$.

proof: Choose $s_0\in S$. s.t. $\overline{s_0} = \overline{x}$ in k. i.e. $x - s_0\in\mathfrak{m}_K = \pi\mathcal{O}_K$. Then write $x = x_0 + \pi x_1$, $x_1\in\mathcal{O}_K$.

Then choose $s_1\in S$. s.t. $\overline{s_1} = \overline{x}$ ... hence $x = s_0 + s_1\pi + \cdots + s_n\pi^n + \pi^{n+1}x_{n+1}$.

Uniqueness: $\sum_{n\geq 0} s_n\pi^n = \sum_{n\geq 0} s_n'\pi^n$, and n be the first index with $s_n\neq s_n'$, then

$(s_n - s_n')\pi^n = \sum_{m>n}(s_m' - s_m)\pi^{m+1}$. where $s_n - s_n'$ is a unit, $v(\text{LHS}) = n < v(\text{RHS})$ $\square$.

Lemma VII.1.5. (Topological Nakayama) Let M be a compact addictive $\mathbb{Z}_p$-module with $\bigcap_{n\geq 0} p^nM = \{0\}$.

If M/pM is spanned over $\mathbb{F}_p$ by the images of $x_1,\ldots,x_N\in M$, then $M = \sum_{i=1}^N \mathbb{Z}_p x_i$.

proof: Let $H = \sum_{i=1}^N \mathbb{Z}_p x_i$. Then H is the continuous image of $\mathbb{Z}_p^N$ (compact)

$\Rightarrow$ H is closed in M

Hence H is compact. By the assumption, $M = H + pM$. (Something we are used to do)

$\Rightarrow M = H + p(H+pM) = H + p^2M = \cdots = H + p^nM$.

Now fix $m\in M$, $(m-H)\cap p^nM$ are nonemty, closed, and we have $(m-H)\cap p^nM \supseteq (m-H)\cap p^{n+1}M$

$\Rightarrow$ By the property of compactness $(m-H)\cap\left(\bigcap_{n\geq 0} p^n M\right)\neq\emptyset$ ($\bigcap_{n\geq 0} p^n M = \{0\}$) $\Rightarrow 0\in m-H \Rightarrow m\in H \Rightarrow M\subseteq H$ $\square$

Back to the proof: $\mathfrak{p}^\circ K$ is non-archimedean.

Case 1: $\operatorname{char} K = 0$

Find $\mathbb{Q}_p$ inside K.

Since $\operatorname{char} K = 0$, we have $\mathbb{Q}\subseteq K$; $\operatorname{char} k = p$, we have $p\in\mathfrak{m}_k \Rightarrow v(p)\geq 1$.

We claim that every nontrivial discrete valuation on $\mathbb{Q}$ is a multiple of v_p

If $l\neq p$ be another prime. $1 = ap+bl$. then $0 = v(1) \geq \min\{\underbrace{v(ap)}_{\geq 1}, \underbrace{v(bl)}_{=0}\}$

$\Rightarrow v(bl)=0 \Rightarrow v(l)=0$ Therefore, $v|_{\mathbb{Q}^\times} = e v_p$, $e := v(p)$.

Since K is complete containing $\overline{\mathbb{Q}}$. By uniqueness of completion: $\overline{\mathbb{Q}}\cong\mathbb{Q}_p$.

Now we prove the foundamental equality: $[K:\mathbb{Q}_p] = ef$. (It's directly from the ramification theory) $\square$. (Here, by lemma 1.5, (take $M=\mathcal{O}_K$), $|\mathcal{O}_K/p\mathcal{O}_K| = |\mathcal{O}_K/\mathfrak{m}_K^e| = |k|^e = p^{ef}$, $\mathcal{O}_K = \sum_{i=1}^{ef}\mathbb{Z}_p x_i$) $\cong \mathbb{Z}_p^{\,n}$

Case 2: $\operatorname{char} K = p$. and $k = \mathcal{O}_K/\mathfrak{m}_K = \mathbb{F}_q$. , $q = p^f$.

Def VII.1.6. (Teichmüller representatives) For $a\in k$, choose $x\in\mathcal{O}_K$ with $\overline{x} = a$. define $\omega(a) = \lim_{n\to\infty} x^{q^n}$.

(Prop). Then $\omega: k\to\mathcal{O}_K$ is a ring homomorphism splitting the residue map: $\mathcal{O}_K\to k$

$\omega(k)$ is called the set of Teichmüller representatives

proof: ① Convergence: It suffices to show that: $x^{q^{n+1}} \equiv x^{q^n} \pmod{\mathfrak{m}_K^{n+1}}$ $\forall n\in\mathbb{N}$.

We prove this by induction on n: Since $\overline{x}^q = \overline{x}$ in $k=\mathbb{F}_q$. $\Rightarrow x^q \equiv x \pmod{\mathfrak{m}_K}$

If $y\equiv z \pmod{\mathfrak{m}_K^n}$., then $y^q \equiv z^q \pmod{\mathfrak{m}_K^{n+1}}$.

Indeed, ~~$y^q = (y-z)^q$ (Undergraduate's dream theorem)~~

$$y^q - z^q = \sum_{i=0}^{q-2}\binom{q}{i} z^i (y-z)^{q-i} + \underbrace{\binom{q}{q-1} z^{q-1}(y-z)}_{=} \equiv 0 \ (\operatorname{char} k = p) \equiv 0 \pmod{\mathfrak{m}_K^{n+1}}$$

② Well-definedness: If $x\equiv y \pmod{\mathfrak{m}_K}$ with $\overline{x} = \overline{y} = a$. Then $x^{q^n} \equiv y^{q^n} \pmod{\mathfrak{m}_K^{n+1}}$

Therefore, $(x^{q^n})_{n\geq 1}$ and $(y^{q^n})_{n\geq 1}$ are Cauchy-equivalent.

③ homomorphism: 1. $\overline{\omega(a)} = a$ (Since $x^{q^n}\equiv x \mod \mathfrak{m}_K$)

2. $\omega(ab) = \omega(a)\omega(b)$ $\lim_{n\to\infty}(xy)^{q^n} = \lim_{n\to\infty} x^{q^n}\lim_{n\to\infty} y^{q^n}$

3. $\omega(a+b) = \omega(a)+\omega(b)$ $\lim_{n\to\infty}(x+y)^{q^n} = \lim_{n\to\infty} x^{q^n} + \lim_{n\to\infty} y^{q^n}$

4. $\omega(1) = 1$, $\omega(0) = 0$ and ω is injective $\square$

Back to Case 2: Apply Lemma VII.1.4 with the representative set $S = \omega(k)$, then $\forall\, x\in\mathcal{O}_K$

$x = \sum_{n\geq 0}\omega(c_n)\pi^n$, $c_n\in k$. Now we define $\Phi: \mathbb{F}_q[[T]]\to\mathcal{O}_K$, $\sum_{n\geq 0} c_n T^n \mapsto \sum_{n\geq 0}\omega(c_n)\pi^n$

Clearly, Φ is bijective. It's a ring homomorphism: Multiplicativity:

$$\left(\sum_{n\geq 0}\omega(c_n)\pi^n\right)\left(\sum_{m\geq 0}\omega(d_m)\pi^m\right) = \sum_{N\geq 0}\left(\sum_{n+m=N}\omega(c_n d_m)\right)\pi^N = \sum_{N\geq 0}\omega\left(\sum_{n+m=N} c_n d_m\right)\pi^N.$$

Moreover. $\Phi(T^n\mathbb{F}_q[[T]]) = \mathfrak{m}_K^n$ So Φ preserves the neighbourhood bases of $0 \Rightarrow \Phi$ is a homeomorphism

$\Rightarrow \mathcal{O}_K \cong \mathbb{F}_q[[T]]$ Moreover, $K \cong \mathbb{F}_q((T))$ (Formal Laurent series field).

2° K is archimedean, then the valuation is equivalent to $|\cdot|$ (Ostrowski) $\Rightarrow K \cong \mathbb{R}$ or $\mathbb{C}$ $\square$.

2. The structure of the unit group of a local field.

Our main goal of this section is to prove :

Thm VII.2.1. Let K be non-archimedean with $k = \mathbb{F}_q$, $q = p^f$. and $\mathcal{O}_K^\times$ be the unit group. $K^\times$ be the multiplicative group. (1) If $\mathrm{char} K = 0$, then $\mathcal{O}_K^\times \cong \mu(K) \times \mathbb{Z}_p^n$, $K^\times \cong \mathbb{Z} \times \mu(K) \times \mathbb{Z}_p^n$ where $n = [K:\mathbb{Q}_p]$; (2) If $\mathrm{char} K = p$, then $\mathcal{O}_K^\times \cong \mathbb{F}_q^\times \times U^{(1)}$, $K^\times \cong \mathbb{Z} \times \mathbb{F}_q^\times \times U^{(1)}$

Def VII.2.2. For $n \geq 0$, let $U^{(0)} := \mathcal{O}_K^\times$, $U^{(n)} := 1 + \mathfrak{m}_K^n$ $(n \geq 1)$ be the unit filtration.

Lemma VII.2.3. $U^{(n)} \leq \mathcal{O}_K^\times$, and we have a decending filtration: $\mathcal{O}_K^\times = U^{(0)} \supseteq U^{(1)} \supseteq \dots$, $\bigcap_{n\geq 0} U^{(n)} = \{1\}$.

proof: (1) From the basic fact from commucative algebra: $\forall x \in \mathfrak{m}_K^n \subseteq \mathfrak{m}_K$, $1+x$ is a unit

$$(1+x)^{-1} = \sum_{j\geq 0}(-x)^j = 1 + \sum_{j\geq 1}(-x)^j \in U^{(n)}, \quad \forall x, y \in \mathfrak{m}_K^n. \quad (1+x)(1+y) \in U^{(n)}$$

Therefore $U^{(n)} \leq U^{(0)}$

(2). If $u \in \bigcap_{n\geq 0} U^{(n)}$, then $u = 1 + x$ where $x = \bigcap_{n\geq 0} \mathfrak{m}_K^n = \{0\}$. So $u = 1$ $\square$.

Lemma VII.2.4. (Quotients) There're isomorphisms of groups: $U^{(0)}/U^{(1)} \cong k^\times$, $U^{(n)}/U^{(n+1)} \cong (k,+)$ $(n \geq 1)$

Consequently, $U^{(n)}/U^{(n+1)} \cong (\mathbb{Z}/p\mathbb{Z})^f$ and $|U^{(1)}/U^{(n)}| = q^{n-1}$ $(n \geq 1)$

proof: (Sketched). (1). $\pi: \mathcal{O}_K^\times \to k^\times$ is surjective. $\ker \pi = U^{(1)}$

(2) $\theta_n: U^{(n)}/U^{(n+1)} \to \mathfrak{m}_K^n/\mathfrak{m}_K^{n+1}$ $(1+x)U^{(n+1)} \mapsto x \bmod \mathfrak{m}_K^{n+1}$ where $x \in \mathfrak{m}_K^n$.

Then $\ker \theta_n = \{0\}$., $\mathfrak{m}_K^n/\mathfrak{m}_K^{n+1} \cong k$ $(x \mapsto \pi^{-n}x \bmod \mathfrak{m}_K)$

Since $k \cong (\mathbb{Z}/p\mathbb{Z})^f \Rightarrow U^{(n)}/U^{(n+1} \cong (k,+) \cong (\mathbb{Z}/p\mathbb{Z})^f$ $\square$.

.emma VII.2.5. (Profinite structure). $\mathcal{O}_K^\times$ is compact and the natural map $\pi: \mathcal{O}_K^\times \to \varprojlim_n \mathcal{O}_K^\times/U^{(n)}$ is an isomorphism of topological groups. In particular, $U^{(1)} \cong \varprojlim_n U^{(1)}/U^{(n)}$

proof: $\mathcal{O}_K^\times = \mathcal{O}_K - \mathfrak{m}_K$. where $\mathfrak{m}_K$ is open. $\mathcal{O}_K$ is compact $\Rightarrow \mathcal{O}_K^\times$ is compact.

Injective: $\ker \pi = \{u \in \mathcal{O}_K^\times \mid u \equiv 1 \pmod{U^{(n)}}, \forall n \geq 0\} = \bigcap_{n\geq 0} U^{(n)} = \{1\}$.

Surjective: $\forall$ compatible system of cosets, choose lifts u_n; then $\frac{u_n}{u_m} \in U^{(m)}$ $(n \geq m)$

$\Rightarrow u_n - u_m = u_m\left(\frac{u_n}{u_m} - 1\right) \in \mathfrak{m}_K^m \Rightarrow (u_n)_{n\geq 0}$ is a Cauchy sequence. ~~and~~

where $u_n \in \mathcal{O}_K^\times$. Since $\mathcal{O}_K^\times$ is closed $\Rightarrow \exists\, u \in \mathcal{O}_K^\times$ s.t. $\pi(u) = \dots$ ▨.

homeomorphism: $\forall$ open neighborhood $U^{(n)} \subseteq \mathcal{O}_K^\times$, $\pi(U^{(n)}) = V_n = \{(a_m)_{m\geq 0} \mid a_0 = 1, \dots, a_{n-1} = 1\}$

is open. The inverse is also right. So it's a homeomorphism $\square$

We first give a rough decomposition:

"Thm VII.2.6. $\mathcal{O}_K^\times \cong k^\times \times U^{(1)} \cong \mu_{q-1}(K) \times U^{(1)}$, $K^\times \cong \mathbb{Z} \times \mathcal{O}_K^\times$ as topological groups, where $\mathbb{Z}$, $k^\times$ are discrete.

Consequently, $K^\times \cong \mathbb{Z} \times \mathbb{F}_q^\times \times U^{(1)}$

proof: We use the Teichmüller lift to prove the thm:

(1) The short exact sequence: $1 \to U^{(1)} \xrightarrow{i} \mathcal{O}_K^\times \xrightarrow{\pi} k^\times \to 1$. is split by the Teichmuller section $\omega: k^\times \to \mathcal{O}_K^\times$ s.t. $\omega \circ \pi = id_{k^\times}$.

Then by Module theory, we have $\mathcal{O}_K^\times \cong k^\times \times U^{(1)}$

(2) Homeomorphism: $\varphi: k^\times \times U^{(1)} \to \mathcal{O}_K^\times$ $(a,u) \mapsto \omega(a)u$ is continuous and ~~sur~~ bijective where $k^\times \times U^{(1)}$ is compact, $\mathcal{O}_K^\times$ is Hausdorff $\Rightarrow$ φ is a homeomorphism.

(3). $\forall x \in K^\times$, $\exists! u \in \mathcal{O}_K^\times$ s.t. $x = \pi^{v(x)} \cdot u$. def: $\psi: K^\times \to \mathbb{Z} \times \mathcal{O}_K^\times$ $x \mapsto (v(x), x\pi^{-v(x)})$

is a homomorphism Homeomorphism: $\psi^{-1}(\{n\} \times \mathcal{O}_K^\times) = \pi^n \mathcal{O}_K^\times$ is open and compact

$\Rightarrow \psi|_{\pi^n \times \mathcal{O}_K^\times}$ is a homeomorphism. And we have $K^\times = \bigsqcup_{n \geq 0} \pi^n \mathcal{O}_K^\times \Rightarrow \psi$ is a homeomorphism □

Lemma VII.2.7. $U^{(1)}$ has no nontrivial torsion of order prime to p.

proof: Let $u = 1 + x \in U^{(1)}$ with $x \neq 0$. Suppose that $u^s = 1$, $(s,p) = 1$

Then $0 = (1+x)^s - 1 = sx + \binom{s}{2}x^2 + \dots + x^s$ Here $v(sx) = v(x)$ since $s \notin \mathfrak{m}_K$. ($\mathrm{char}\, k = p$)

and $v\left(\binom{s}{j}x^j\right) \geq j\,v(x) > v(x) \Rightarrow v(x) = 0$, $x \in \mathfrak{m}_K$ Contradiction! □.

Rmk. We now give an important view of point: Every compact abelian pro-p group G is a $\mathbb{Z}_p$-module: For $u \in G$, and $z \in \mathbb{Z}_p$, choose $m_i \in \mathbb{Z}$ s.t. $m_i \to z$ (in p-adic sense)

and define $u^z := \lim_{i \to \infty} u^{m_i}$ The limit exists and is independent of $(m_i)_{i \geq 1}$.

Now we come back to our main theorem:

1°. If $\mathrm{char}\, K = 0$. We need two operators: the logarithm and the expontential.

Def VII.2.8. For $x \in \mathfrak{m}_K$, let $\log(1+x) := \sum_{k \geq 1} (-1)^{k+1} \frac{x^{k+1}}{k}$, $\exp(x) := \sum_{k \geq 0} \frac{x^k}{k!}$

Prop VII.2.9. (1) $\log(1+x)$ converges for every $x \in \mathfrak{m}_K$, and defines a continuous homomorphism

$\log: U^{(1)} \to (\mathfrak{m}_K, +)$

(2) $\exp(x)$ converges for every x with $v(x) > \frac{e}{p-1}$.

(3) If $m \in \mathbb{Z}$ with $m > \frac{e}{p-1}$, then log and exp are mutually inverse isomorphisms

$\log: U^{(m)} \to \mathfrak{m}_K^m$ $\exp: \mathfrak{m}_K^m \to U^{(m)}$

Moreover, they induce the identity on $U^{(m)}/U^{(m+1)} \cong \mathfrak{m}_K^m/\mathfrak{m}_K^{m+1} \cong k$.

proof: (1). For $x \in \mathfrak{m}_K$, and $k \geq 1$, $v\left(\frac{x^k}{k}\right) = k v(x) - e v_p(k) \geq k \underbrace{v(x)}_{\geq 1} - e \log_p k \to +\infty$ (Here we use $v_p(k) \leq \log_p k$)

Hence the series converges. $\log((1+X)(1+Y)) = \log(1+X) + \log(1+Y)$ holds in $\mathbb{Q}[[X, Y]]$,

substituting $X = x$, $Y = y$ gives $\log(uv) = \log u + \log v$. for $u, v \in U^{(1)}$

(2) For $v(x) > \frac{e}{p-1}$. and $k \geq 1$. $v\left(\frac{x^k}{k!}\right) = k v(x) - e v_p(k!) \geq k v(x) - \frac{e(k-1)}{p-1} \to +\infty$ ($v_p(k!) = \frac{k - S_p(k)}{p-1}$)

So $\exp(x)$ converges.

(3) Let $m > \frac{e}{p-1}$, $x \in \mathfrak{m}_K^m \Rightarrow v\left(\frac{x^k}{k}\right) \geq km - \frac{e(k-1)}{p-1} = m + (k-1)\left(m - \frac{e}{p-1}\right) > m$.

Similarly for $v\left(\frac{x^k}{k!}\right)$, then $\log(1+x) \equiv x \pmod{\mathfrak{m}_K^{m+1}}$, $\exp(x) \equiv 1 + x \pmod{\mathfrak{m}_K^{m+1}}$.

Therefore log maps $U^{(m)}$ into $\mathfrak{m}_K^m$, ... for exp

The identity ~~holds~~ $\log(\exp(X)) = X$, $\exp(\log(X)) = X$ holds in $\mathbb{Q}[[X]]$, hence in K □

ThmVII.2.10. Char $K = 0$, then $U^{(1)} \cong \mu_{pa}(K) \times \mathbb{Z}_p^n$, Together with ThmVII.2.6, we have

$$O_K^\times \cong \mu_{q-1}(K) \times \mu_{pa}(K) \times \mathbb{Z}_p^n, \quad K^\times \cong \mathbb{Z} \times \mu(K) \times \mathbb{Z}_p^n. \quad \text{(Exciting !!!)}$$
$$\cong \mu(K) \times \mathbb{Z}_p^n$$

proof:

Lemma VII.2.11. $(\mathrm{Tor}\, U^{(1)})$: $\mathrm{Tor}\, U^{(1)} = \mu_{p^\infty}(K) := \bigcup_{r \geq 1} \mu_{p^r}(K) = \mu_{pa}(K)$

pf: (sketch) Use Lemma VII.2.7. □.

Step 1. Choose $m > \frac{e}{p-1}$, $e = v(p)$. By (2.9) $U^{(m)} \cong \mathfrak{m}_K^m$ and $(\mathfrak{m}_K^m, +) \cong (O_K, +)$ $(x \mapsto \pi^{-m} x)$

By (1.3). $(O_K, +) \cong \mathbb{Z}_p^n$. Hence $U^{(m)} \cong \mathbb{Z}_p^n$.

Step 2. Since $U^{(m)}$ is torsion-free, we have $\overline{U}^{(m)} \cong \mathbb{Z}_p^n$ in $U^{(1)}/\mu_{pa}(K)$.

Moreover, $[U^{(1)}/\mu_{pa}(K) : U^{(m)}] \leq [U^{(1)} : U^{(m)}] < \infty \Rightarrow U^{(m)}$ is open with finite index.

Step 3. $M = U^{(1)}/\mu_{pa}(K)$ by Rmk, we know that M is a compact $\mathbb{Z}_p$-module.

It's easy to check that M is torsion free, finitely-generated module over $\mathbb{Z}_p$ (PID).

Hence M is a free module! $\Rightarrow M \cong \mathbb{Z}_p^n$.

Step 4. Now we are familar with the following sequence: $1 \to \mu_{pa}(K) \to U^{(1)} \to \mathbb{Z}_p^n \to 1$.

Since $\mathbb{Z}_p^n$ free, so it splits, $\Rightarrow U^{(1)} \cong \mu_{pa}(K) \times \mathbb{Z}_p^n$ □.

◦ If char $K = p$. W.L.O.G Let $K = \mathbb{F}_q((T))$, $O_K = \mathbb{F}_q[[T]]$., $\pi = T$.

ThmVII.2.12. Char $K = p$, $q = p^f = |k|$, Then $O_K^\times \cong \mathbb{F}_q^\times \times U^{(1)}$, $K^\times \cong \mathbb{Z} \times \mathbb{F}_q^\times \times U^{(1)}$

where $U^{(1)} = 1 + T\mathbb{F}_q[[T]]$ is a torsion-free abelian pro-p group which is not topologically generated!

proof: (1) $U^{(1)}$ is torsion-free: let $u = 1 + x$, $x \in T\mathbb{F}_q[[T]]$, then by lemma 2.7. $u^{p^r} = 1$ (If u is a torsion)

However, char $K = p \Rightarrow 1 = u^{p^r} = (1+x)^{p^r} = 1 + x^{p^r} \Rightarrow x = 0$, $u = 1$.

(2) Let $U^{(1)p} = \{u^p \mid u \in U^{(1)}\}$ Since $(1+x)^p = 1 + x^p$ We claim that the classes of $1 + T^n$ ($p \nmid n$) are linearly independent over $\mathbb{F}_p$ in $U^{(1)}/U^{(1)p}$. In $U^{(1)p}$, $\exists\, 1 + \sum_{n \geq 1} a_n T^n$. $a_n \neq 0 \Leftrightarrow p \mid a_n$ (*)

Indeed, if $\prod_{p \nmid n} (1+T^n)^{\alpha_n} \in U^{(1)p}$ $\alpha_n \in \mathbb{F}_p$

Let n_0 be the smallest n with $\alpha_n \neq 0$. the coefficient of T^{n_0} is $\alpha_{n_0} \neq 0$. But $p \nmid n_0$

Contradiction! So $\{1+T^n (p+n)\}$ is linearly indapondent over $\mathbb{F}_p$

Now suppose that $U^{(1)}$ are topologically generated by $g_1, \dots, g_r$

Then by definition: $\exists\ \psi: \mathbb{Z}_p{}^r \to U^{(1)}$ $(z_1, \dots, z_r) \mapsto g_1{}^{z_1} \dots g_r{}^{z_r}$ surjective, continuous.

(Rmk. $g_1 \dots g_r$ topologically generate $U^{(1)} \iff \overline{\langle g_1, \dots, g_r \rangle} = U^{(1)}$).

$\Rightarrow U^{(1)}/U^{(1)p}$ is genorated by g_i ($z_i = m + p z_i' \Rightarrow g_i{}^{z_i} \equiv g_i{}^{\overline{z_i}} \pmod{U^{(1)p}}$ where $\overline{z_i} = m$) Contradiction! □

Examples: 1 $K = \mathbb{Q}_p(\zeta_p)\mathbb{Q}_p{}^\times \cong \mathbb{Z} \times \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}_p \Rightarrow \mathbb{Q}_p{}^\times = \langle p \rangle \times \mu_{p-1} \times (1+p\mathbb{Z}_p)$

2. $K = \mathbb{Q}_2$. $\mathbb{Q}_2{}^\times \cong \mathbb{Z} \times \{\pm 1\} \times \mathbb{Z}_2 \Rightarrow \mathbb{Q}_2{}^\times = \langle 2 \rangle \times \{\pm 1\} \times (1+4\mathbb{Z}_2)$.

3. $K = \mathbb{Q}_p(\zeta_p{}^r), (2+p)$, $\mathcal{O}_K{}^\times \cong \mu_{p-1} \times \mu_{p^r} \times \mathbb{Z}_p{}^{\varphi(p^r)}$

4. $K = \mathbb{F}_q((T))$, $K^\times = \langle T \rangle \times \mathbb{F}_q{}^\times \times (1 + T\mathbb{F}_q[[T]])$ □

3. Ramification theory of local fields. (L/K be Galois and K non-archimedean, local; so does L).

Some review of ramification: Let $G = \mathrm{Gal}(L/K)$

Def VII.3.1. (1) The ramification index is $e = v_L(\pi_K)$.

(2) The residue degree $f = [l : k]$ where $l = \mathcal{O}_L/\mathfrak{m}_L$; $k = \mathcal{O}_K/\mathfrak{m}_K$.

(3) L/K is unramified if $e = 1$.

(4) L/K is totally ramified if $f = 1$.

(5) L/K is <u>tamely</u> ramified (弱分歧) if $p \nmid e$

(6) L/K is <u>wildly</u> ramified if $p \mid e$

Rmk. (1) The foundamental equality will be very concise: $[L:K] = ef$.

(2) $1 \to G_0 \to G \to \mathrm{Gal}(l/k) \to 1$ is exact, where $G_0 := \{\sigma \in G \mid \sigma(x) \equiv x \pmod{\mathfrak{m}_L}\ \forall x \in \mathcal{O}_L\}$ is the ~~Inertia~~ Inertia group. The corresponding coset G/G_0 is called <u>the Frobenius</u>. If L/K is unramified, the Frobenius is a unique element $\mathrm{Frob}_{L/K} \in G$ which generates G. ($G/G_0 \cong \mathrm{Gal}(l/k)$)

In particular, $|G/G_0| = f$, $|G_0| = e$.

Recall that in global field: $1 \to I_\beta \to D_\beta \to \mathrm{Gal}(l/k) \to 1$ exact, I_β is the Inertia group. D_β is the decomposition group and $|\mathrm{Gal}(l/k)| = f$, $|I_\beta| = e$, $|D_\beta| = e \cdot f$. Now $D_\beta = \mathrm{Gal}(L/K)$

Def VII.3.2. (Ramification groups) For $i \geq -1$, let $G_i := \{\sigma \in G \mid v_L(\sigma(x) - x) \geq i+1$, for all $x \in \mathcal{O}_L\}$.

Then $G_{-1} = G$, G_0 is the inertia group, G_1 is the wild ramification group.

Lemma VII.3.3. For $\sigma \in G_0$ and $i \geq 0$, TFAE: (1) $\sigma \in G_i$; (2) $v_L(\sigma(\pi_L) - \pi_L) \geq i+1$; (3) $\sigma(u) \equiv u \pmod{\mathfrak{m}_L{}^{i+1}}$, $u \in \mathcal{O}_L{}^\times$.

proof: (1) $\Rightarrow$ (3): Obvious, since $\sigma(u) - u = u\left(\frac{\sigma(u)}{u} - 1\right)$

(3) $\Rightarrow$ (2): $u = 1 + \pi_L \Rightarrow \sigma(1+\pi_L) \equiv 1 + \pi_L \pmod{\mathfrak{m}_L{}^{i+1}} \Rightarrow \sigma(\pi_L) \equiv \pi_L \pmod{\mathfrak{m}_L{}^{i+1}}$

(2) $\Rightarrow$ (1): $\forall x \in \mathcal{O}_L$, $x = \sum_{j \geq 0} c_j \pi_L{}^j$ where c_j is the Teichmuller representative,

$\forall \sigma \in G_0$, $\sigma(c_j) = c_j$ (Since $\sigma(l) = l$) $\Rightarrow \sigma(x) - x = \sum_{j \geq 0} c_j (\sigma(\pi_L)^j - \pi_L{}^j) \equiv 0 \pmod{\mathfrak{m}$

rop VII.3.3. 1. $G_i \trianglelefteq G$, $G = G_{-1} \supseteq G_0 \supseteq G_1 \supseteq$

2. $G_i = \{1\}$ for sufficiently large i.

3. G_0/G_1 is cyclic of order prime to p, G_i/G_{i+1} $(i \geq 1)$ is abelian p-group.

Consequently, G_1 is the unique p-Sylow subgroup of G_0, and G is soluable.

proof: (1) For $\tau \in G$ and $\sigma \in G_i$, $v_L(\tau\sigma\tau^{-1}(x) - x) = v_L\left(\tau\left(\sigma\tau^{-1}(x) - \tau^{-1}(x)\right)\right) = v_L\left(\sigma\tau^{-1}(x) - \tau^{-1}(x)\right) \geq i+1$

$\Rightarrow G_i \trianglelefteq G$

(2). Choose $\alpha \in \mathcal{O}_L$ with $L = K(\alpha)$. If $\sigma \neq 1$, then $\sigma(\alpha) \neq \alpha$, so $v_L(\sigma(\alpha) - \alpha) = m$ [illegible], then $\sigma \notin G_i$ for $i \geq m$. Notice that $|G|$ is finite.

(3). For $\sigma \in G_0$, write $\sigma(\pi_L) = u_\sigma \cdot \pi_L$ where $u_\sigma \in \mathcal{O}_L^\times$, then $u_{\sigma\tau} = \sigma(u_\tau)u_\sigma$ and $\sigma(u_\tau) \equiv u_\tau \pmod{\mathfrak{m}_L}$

We have $\varphi: G_0 \hookrightarrow l^\times$, $\sigma \to u_\sigma \pmod{\mathfrak{m}_L}$ is an injective homomorphism, $\ker\varphi = G_1$

hence G_0/G_1 is cyclic of order dividing $|l^\times| = q^f - 1$ prime to p.

For $i \geq 1$, consider $\psi: G_i \hookrightarrow \mathfrak{m}_L^{i+1}/\mathfrak{m}_L^{i+2} \cong (l, +)$ $\sigma \mapsto \sigma(\pi_L) - \pi_L \pmod{\mathfrak{m}_L^{i+2}}$

$\sigma\tau(\pi_L) - \pi_L = (\sigma(\pi_L) - \pi_L) + \sigma(\tau(\pi_L) - \pi_L) \Rightarrow \psi(\sigma\tau) = \psi(\sigma) + \psi(\tau)$ (homomorphism)

[illegible] $\ker\psi = G_{i+1}$ $\Rightarrow$ G_i/G_{i+1} embeds into $(l, +) \cong (\mathbb{Z}/p\mathbb{Z})^f$.

$\Rightarrow$ G is solvable $\square$.

Coro VII.3.4. L/K is tamely ramifield $\Leftrightarrow$ $p \nmid e$ $\Leftrightarrow$ $G_1 = \{1\}$. (Since $e = |G_0| = |G_0/G_1| \cdot |G_1|$).

Prop. VII.3.5. Let $K_{ur} := L^{G_0}$ and $K_t := L^{G_1}$ where $L^{G_0} = \{x \in L \mid \sigma(x) = x, \forall \sigma \in G_0\}$ ...

Then (1) K_{ur}/K is the maximal unramified subextension of L/K with $[K_{ur} : K] = f$, L/K_{ur} is totally ramified with index e

(2) K_t/K is the maximal tamely ramified subextension of L/K;

(3) $K \xrightarrow{f} K_{ur} \xrightarrow{e_0} K_t \xrightarrow{p^{v_p(e)}} L$, $G \supseteq G_0 \supseteq G_1 \supseteq \{1\}$. with G/G_0 cyclic (Frobenius)
$|G_0/G_1| = e_0$, $|G_1| = p^{v_p(e)}$

(4) Moreover, the unramified extension of K of degree f is unique: it's $K(\mu_{q^f-1})$.

proof: (1) $G \to \mathrm{Gal}(l/k)$ has kernel G_0. Then by Galois's theory, $\mathrm{Gal}(K_{ur}/K) \cong G/G_0 \cong \mathrm{Gal}(l/k)$ and $f(K_{ur}/K) = f$.

(2) (3) are obvious from (1)

(4)

Now we give a beautiful and elegant theorem:

Thm VII.3.6. Let L/K be totally ramified of degree e with $p \nmid e$. Then:

(1) $\exists\, \pi_L$ s.t. π_L^e is a uniformizer of K; consequently $L = K(\pi_L)$

(2) L/K is Galois iff $\mu_e \subseteq K$; in that case L/K is cyclic and $\mathrm{Gal}(L/K) \cong \mu_e \cong \mathbb{Z}/e\mathbb{Z}$ via $\sigma \mapsto \sigma(\pi_L)/\pi_L$